
Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

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Abstract

1 Quantizing a large language model reduces serving cost and costs some accu-
2 racy. Determining how much normally requires re-running a full benchmark suite,
3 which is the expense quantization was meant to avoid. We ask whether the accu-
4 racy delta can be predicted instead. Using 817 published evaluations spanning 38
5 base checkpoints, 8 model families and 6 quantization schemes, we report three
6 results. First, a *negative* one: per-model prediction carries almost no signal. Un-
7 der leave-one-family-out validation the best predictor, a per-scheme mean, beats
8 a global mean by 0.03pp of MAE (95% checkpoint-bootstrap CI [0.01, 0.05]),
9 and both ridge regression and gradient boosting are *worse* than guessing the av-
10 erage, because evaluation noise accounts for at least a third of the variance on 8
11 of 11 benchmarks. Second, a positive one: conditioning only on the quantization
12 scheme yields per-scheme conformal-style intervals with 90.1% empirical cover-
13 age (118/131, 95% CI [83.6, 94.6]) on checkpoints never used in construction.
14 Those checkpoints cover 4 of the 6 schemes, and W4A16 supplies 80 of the 131
15 rows. Each (scheme, size-band) cell returns one of three verdicts: trusted; insuf-
16 ficient evidence, where the interval is still shown but the scheme is demoted and
17 flagged; or refused, where the interval is withheld. Currently 9 of 17 cells are in-
18 sufficient evidence and 0 are refused. Third, and most consequential for practice:
19 the published corpus is outcome-truncated. Every configuration in it is one that
20 someone chose to release. We rented an A100 and measured deliberately-faulted
21 configurations, observing losses to -39.5pp against a published worst case of
22 -8.86pp —measured, we stress, only on models of 1.5B and below, and on config-
23 urations engineered to fail rather than typical ones. This is an existence proof that
24 far worse outcomes exist than the corpus contains; it does not estimate how com-
25 mon they are. We argue the resulting artifact should be described as a calibrated
26 historical baseline rather than a predictor, and we state precisely why its coverage
27 guarantee does not transfer to a practitioner’s own untuned recipe. Code, data and
28 the provenance of every reported figure are provided as supplementary material.

29 1 Introduction

30 Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8
31 wants one number: how much accuracy will this cost? In practice the only way to know is to run the
32 evaluation.

33 This paper asks whether that number can be forecast from the public record. Model publishers
34 release quantized checkpoints with benchmark tables comparing the compressed model against its
35 dense parent. Aggregated, these are hundreds of paired measurements of exactly the quantity of
36 interest (850 extracted in our harvest).

37 Our contributions:

- 38 1. **A negative result** (§5). Per-model prediction of the accuracy delta does not work, and
39 we show why: the target is dominated by evaluation noise. We report this as the primary
40 finding rather than burying it.
- 41 2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured
42 coverage on held-out families and a three-state verdict for each (scheme, size-band) cell:
43 *trusted*; *insufficient evidence*, where the interval is still shown but the scheme is demoted
44 and the cell flagged as one the tool cannot judge; or *refused*, where the interval is withheld.
45 Currently 9 of 17 cells are insufficient evidence and 0 are refused.
- 46 3. **An existence proof that the corpus omits severe outcomes** (§7), via GPU experiments on
47 deliberately-faulted configurations—data the published record structurally cannot contain.
48 It shows that losses far beyond anything published are reachable on small models. It does
49 not estimate how large the bias is: our attempt at a correction (in the supplement) did not
50 work well enough to ship.
- 51 4. **An audit** (§9) recording errors we found in our own work, including ones that changed
52 published numbers and claims.

53 2 Related Work

54 **Predicting evaluations.** BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism
55 closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and
56 decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks
57 is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is
58 a model’s absolute score; ours is the paired delta between a model and its compressed self, and
59 quantized checkpoints do not appear in their paper.

60 **Conformal prediction on compressed models.** Tong et al. [2026] apply conformal prediction
61 directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand
62 differs: they evaluate predictive uncertainty (coverage and set size) *after* running the compressed
63 model. Ours is constructed over historical accuracy deltas to give an interval *before* running any-
64 thing. Their Figure 2 plots the same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it.
65 Their finding that compression decouples accuracy from uncertainty bounds what an accuracy-only
66 estimate like ours can claim (§8).

67 **Quantization methods.** The checkpoints we harvest are produced with GPTQ Frantar et al. [2022]
68 and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024]. Kumar et al. [2024]
69 fit a functional form that predicts degradation from training and inference in varied precisions, using
70 pretraining runs of models up to 1.7B parameters. We fit no such law: our envelope is empirical, per
71 scheme, over published post-training-quantized checkpoints.

72 **The source of our data.** Kurtić et al. [2024] report over 500,000 evaluations of quantized Llama-
73 3.1 models, finding recovery above 99% of the average OpenLLM v1 score for every scheme tested.
74 As far as we can tell, that work does not address whether the published set of quantized models is a
75 biased sample. Characterising that gap is the contribution of §7.

76 **Selection bias.** Survivorship bias has been identified and measured in a widely used retrieval
77 benchmark Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to fail un-
78 der selection: conditional on a selection event, calibration data are no longer exchangeable with a
79 test point Barber et al. [2023], Jin and Ren [2024]. We apply this directly to our own artifact in §8.

80 3 Data

81 We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yield-
82 ing 850 rows, of which 817 remain after dropping 33 whose baseline accuracy is below 20% (near
83 chance, where a delta is uninformative), across 38 base checkpoints, 8 families and 6 schemes.

84 **An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair.
 85 This is redundant information, and we exploit it: for every three-column row we check recovery $\approx$
 86 $100 \times \text{after/before}$ to a tolerance derived from the card’s own printed precision, and reject rows that
 87 disagree. This caught spurious values that our parser produced under column-order variation be-
 88 tween card templates, and internally inconsistent cards. Rejections are recorded rather than silently
 89 dropped.

90 **A structural limitation, stated at the outset.** All rows come from a single publisher using a
 91 single toolchain. Holding out a model family tests transfer across *models*; nothing in this design
 92 tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

93 **The pathway is only sparsely documented.** A live scan on 2026-09-26 across the 439 RedHatAI
 94 quantized-model cards currently published on Hugging Face (of which 102 contribute rows to our
 95 harvested dataset) shows how sparsely the pathway that produced these outcomes is documented:
 96 12 of 439 cards pin a quantization-library version, 33 mention activation reordering, 17 mention
 97 dampening, and 198 of 439 report none of `num_calibration_samples`, `max_sequence_length`,
 98 activation reordering or dampening. On the 102-card subset that our dataset draws rows from, 0
 99 pin a library version and 39 of 102 report none of those four parameters. (This RedHatAI-only
 100 scan is distinct from the 10-publisher, 260-card publisher census reported in §9, which is an earlier
 101 stable snapshot spanning multiple publishers and asks a different question.) The published record we
 102 calibrate against is therefore a record of outcomes with the pathway that produced them substantially
 103 undocumented on the publisher whose cards contribute the largest single share of our corpus, which
 104 is one of the reasons an outcome-truncation argument (§7) cannot be closed from the corpus alone.

105 4 Method

106 **Conformal-style intervals.** We use the residual-quantile construction of split conformal re-
 107 gression Lei et al. [2018]: given residuals $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the
 108 $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which under a proper calibration/test split gives exact finite-
 109 sample coverage under exchangeability. Where n is too small for that index to exist, we return
 110 ∞ rather than a misleadingly finite band. The shipped per-scheme interval fits its centre and its
 111 half-width on the *same* rows (an in-sample residual-quantile band), so the split-conformal guaran-
 112 tee holds only for the explicit calibration-set variant used inside the leave-one-family-out audit; the
 113 empirical prospective figure below is what carries the shipped intervals.

114 **Mondrian (group-conditional) calibration.** Calibrating separately within categories is the Mon-
 115 drian construction of Vovk et al. [2005], applied to regression by Boström and Johansson [2020]. A
 116 single marginal interval advertised at 90% covers only 62% of NVFP4 rows under leave-one-family-
 117 out (recomputed live)—the users who most need it. We therefore calibrate per scheme.

118 **A three-state verdict.** For each (scheme, size-band) cell we measure coverage directly, and the
 119 tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and ad-
 120 equately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is
 121 measurably poor: the checkpoint-bootstrap interval on its coverage lies wholly below the 85% line,
 122 on at least 3 checkpoints. It is *insufficient evidence* when the cell rests on fewer than 3 distinct
 123 checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on its coverage straddles the
 124 85% line. In that case the evidence cannot support either of the other two verdicts. The tool still
 125 prints the all-sizes interval for such a cell, but demotes the scheme to the lowest tier and states that
 126 the cell cannot be judged; only a refused cell has its interval withheld.

127 Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound.
 128 The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one
 129 to anything, so counting that as a failure would flag cells that are merely outperforming. Comparing
 130 a one-sided measurement against the printed 90% is the *lenient* reading, not a flattering one: the
 131 interval is symmetric, so the permitted 10% is split across two tails, and pooled one-sided coverage,
 132 across all 817 rows with each row scored under 7 different calibration families, is 95.1%.

133 Currently 0 of 17 cells are refused. Two cells were flagged under the earlier two-state rule, and nei-
 134 ther is a confident refusal under the three-state one. `w4a16|<2B` had one-sided coverage of 70.1%

over 11 rows (two-sided 68.8%), but those rows come from only 2 checkpoints. w8a16|>10B had 81.1% over 31 rows from 5 checkpoints (two-sided 73.7%), and its bootstrap interval, [47, 100]%, straddles the threshold. Both are now *insufficient evidence*. We had scoped the change to those two cells; applying the rule to every cell moved 9 of 17 into that state, including several whose point-estimate coverage looked comfortable. The same measurement and the same rule judge each scheme’s all-sizes coverage: 1 of the 6 schemes is insufficient evidence at that level and 0 is refused. The one-sided scoring and the three-state rule were proposed by an external commenter who identifies as an AI account collaborating with the OpenLanguageModel maintainers; we recomputed each of their figures from the raw rows before adopting either change, and one of their descriptions did not survive that check.

A support floor. Coverage measured on many benchmarks from one checkpoint is not independent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

5 The negative result

Under leave-one-family-out validation, row-weighted MAE:

	predictor	MAE (pp)	beats global mean?
	per-scheme mean (shipped)	0.7227	yes
	global mean (baseline)	0.7545	—
150	per-(scheme $\times$ benchmark)	0.7607	no
	ridge regression	0.7639	no
	per-benchmark mean	0.7647	no
	gradient boosting	0.7878	no

The per-scheme mean’s advantage is small: 0.03pp, with a checkpoint-bootstrap 95% CI of [0.01, 0.05]pp, and it is positive in 7 of the 8 held-out families. It is a distinguishable but modest gain. Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size, bit-width, method, benchmark identity and headroom each degraded out-of-family performance.

Why. Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta should be approximately zero—gives a noise floor that accounts for anywhere from 15% of the observed variance to all of it across 11 benchmarks. On GSM8K the deltas of supposedly lossless schemes have a standard deviation of 2.1pp, while the quantity being predicted has mean -0.39 pp. The target is a difference of two noisy quantities that largely cancel: the numerically smallest and hardest-to-predict summary of quantization damage available. Two cautions on the noise figures. They assume near-lossless schemes have a true delta near zero, so they are upper bounds on noise. And 5 of the 11 estimates rest on 10 or fewer rows; one, at 124%, exceeds the total variance, which we attribute to sampling error rather than treat as a finding.

Implication. The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band, with no model and no conformal machinery, is a serious competitor: it looked better covered in-sample, though under the strict protocol that advantage disappears (§9). What conformal contributes is a finite-sample guarantee and a principled rule for small- n schemes, not accuracy. We therefore describe the result as a *calibrated historical baseline*, not a predictor.

6 Results

	scheme	90% interval (pp)	worst observed	% losing >3pp	evals
	fp8	[-1.11, +0.48]	-1.90pp	0.0%	80
	w8a8_int	[-1.71, +1.07]	-3.32pp	1.0%	194
170	fp8_dynamic	[-1.44, +1.34]	-8.72pp	1.0%	200
	w8a16	[-0.98, +0.81]	-3.04pp	1.2%	86
	w4a16	[-2.87, +1.40]	-8.86pp	5.7%	193
	nvfp4	[-4.10, +1.87]	-8.12pp	15.6%	64

171 **Prospective validation.** After freezing the method we harvested checkpoints not used during de-
 172 velopment. Under a strict protocol that excludes any family present in training and any card whose
 173 format informed a parser fix, coverage is $118/131 = 90.1\%$, 95% Clopper–Pearson CI [83.6, 94.6],
 174 against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code
 175 with the pipeline reproduces this figure with zero disagreements. That checks the computation, not
 176 the validity of the design.

177 **What the prospective set does and does not cover.** The 131 strict rows come from 19 quantized
 178 checkpoints, not 131 independent observations: a bootstrap that resamples checkpoints gives a 95%
 179 interval of [83.2, 94.7]%, close to the row-level one. The rows cover only 4 of the 6 schemes; FP8
 180 and NVFP4 have no prospective rows, so the figure says nothing about them.

	scheme	strict rows	checkpoints	covered (%)
181	w4a16	80	11	87.5
	w8a8_int	36	5	91.7
	fp8_dynamic	11	2	100.0
	w8a16	4	1	100.0

182 W4A16, the most common scheme, dominates the pooled figure and sits below the nominal 90%;
 183 the other schemes rest on few checkpoints. Over all 186 harvested prospective rows, before the strict
 184 filter, coverage is 90.3%, so the filter did not manufacture the headline.

185 **What the envelope cannot do.** 0 of 6 intervals exclude zero: the tool can never state that a scheme
 186 *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when
 187 real damage occurs, the interval usually misses it. This is the single most important caveat in the
 188 paper.

189 7 The measured left tail

190 The corpus contains only configurations someone chose to publish. To measure what it structurally
 191 cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

192 **Result.** Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the pub-
 193 lished corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-
 194 configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor
 195 instead of per-group.

196 **Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes
 197 that catastrophic damage is reachable; it is not a measured bound for larger models.

198 **A confound we found and did not resolve.** Re-running MMLU under the published protocol
 199 (5-shot, letter-scored) recovers accuracies in the range published on the cards: our Qwen2.5-0.5B-
 200 Instruct scores 47.20 against the Instruct card’s 46.83 (the base-model card gives 47.57), and our
 201 Qwen2.5-1.5B-Instruct scores 59.57. The harness question closed. Our control run then compared
 202 Instruct-model checkpoints against Red Hat’s *base-model* W4A16 publications, because no Instruct
 203 W4A16 checkpoints are published: our correctly-configured Instruct control is $2.6\times$ more damag-
 204 ing than the base-model published W4A16 (mean gap -3.78pp , $z = 2.9$ and 5.3). This is not like-
 205 for-like—Instruct models could plausibly be more fragile under 4-bit quantization than their base
 206 counterparts—so the ratio is indicative rather than a clean estimate. A matched Instruct-vs-Instruct
 207 control is deferred future work. The pattern we observed is consistent with approximately $171\times$ less
 208 calibration data than production defaults, though the attribution is not independently tested. Conse-
 209 quently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It
 210 is an indicative anchor, not a bound, and we decline to publish a corrected point estimate: the cor-
 211 rection moves by 5.70pp depending on the anchor set, and we have only 6 rows from two models.

8 Limitations

The guarantee is conditional, and absent where it matters most. Split conformal gives exact coverage under exchangeability between calibration and test. Our calibration set is the set of configurations a publisher chose to release—a selection event. Conditional on selection, calibration points are not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al. [2023], Jin and Ren [2024]. It holds for a new checkpoint drawn from the same publication process. It carries **no guarantee** for a recipe the reader tuned themselves and that no publisher would have released—which is the case a practitioner most wants covered.

Accuracy is not the whole of deployment risk. Tong et al. [2026] show compression frequently decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none of that.

One publisher, one toolchain, and why. See §3. Our own control arm (§7) is direct evidence that practitioner and calibration volume can dominate scheme choice. We measured how confining this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9 publishers carry paired before/after evaluations, 9 of them Intel’s, so others do publish them. What the other publishers almost never provide is a *Recovery* column, which is what lets each row be checked against $100 \times$ after/before and rejected on disagreement: of their rows, 6 pass that gate, all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones published in a self-verifying form.

What can and cannot be re-run. The supplement recomputes every tagged figure from the data files it ships. It cannot re-run the faulted-configuration experiments (they need a GPU; the recorded results are included), the harvest from raw model cards (the cards are not redistributed), or the prospective-card checks (those files are not included), and re-running the low-rank comparison mentioned in the supplement needs a checkout of BenchPress’s released code.

An attempt at re-running the calibration-data test. We attempted to reproduce Red Hat’s published W4A16 checkpoints for two of the corpus’s most-measured base models, Qwen2.5-0.5B and Qwen2.5-1.5B, using the *llm-compressor* recipe at its documented defaults, in order to test whether the roughly $171 \times$ calibration-data ratio between our own control and production defaults accounts for the observed excess damage. Neither of these two cards pins a library version or names a calibration dataset; both record only the resulting weight representation (symmetric per-group INT4, group size 128 on 1.5B and group size 64 on 0.5B).

Rows are not independent. Up to sixteen benchmark rows come from a single quantization run. We mitigate with group-conditional calibration and the support floor, and note that a bootstrap over the 19 quantized checkpoints gives $[83.2, 94.7]\%$, close to the row-level interval, but the unit of analysis remains a row. The leave-family-out coverage measurement has the same dependence: each row is scored once under each of 7 calibration families, so the 5719 evaluations are not independent, and every coverage judgment in this paper resamples whole checkpoints rather than counting evaluations.

9 Audit

We record errors found in our own work, including those that changed published claims. This section exists because the alternative—reporting only what survived—would misrepresent how the result was obtained.

Errors that changed a published number.

- *A retracted headline.* An early claim that held-out models came from families never used in construction was wrong; they were new checkpoints of families already present. The strict figure is 118/131.

- *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was matched as Math-Lv1-5, contaminating 7 test rows with scores from a different benchmark.
- *A centring mismatch.* Calibration measured residuals about the stratum mean while intervals centred on the scheme mean, mis-sizing every calibrated width. Found only by independent reimplementations.
- *A column-order fault.* One card template orders columns [recovery, base, quant], which our parser turned into a spurious -68pp delta. The integrity gate rejected 10 of 11 rows on that card, passing only the one where base = quant made the check degenerate.
- *A retracted single-publisher claim.* We had asserted that one publisher was close to the only source of paired evaluations. A census of 10 publishers found 10 paired-evaluation cards outside the control publisher, 9 of them Intel's; what differs is whose rows can be verified (§8).
- *A bootstrap on the wrong statistic.* The checkpoint-level bootstrap in our coverage audit re-sampled the two-sided indicator while the verdict was judged on the one-sided statistic. An external commenter asked for it to wrap the one-sided figure, which exposed the mismatch (§4).
- *Inflated pair counts.* An earlier version of the tool reported the number of (test row, calibration family) evaluations as if it were the number of rows: with 8 families each row is scored 7 times, so 5719 evaluations came from only 817 rows. Every row-count claim was recomputed on `distinct_rows`.
- *Two levels judged differently.* Cells were judged on the one-sided coverage statistic and refused when the point estimate fell below the threshold, but schemes had been judged on the two-sided statistic and refused only when the entire bootstrap interval fell below it. A single classifier now judges both levels the same way, and the envelope's refused list is a computed copy of the ranking tool's, checked by a test.

A regression we nearly shipped. An audit indicated that a raw asymmetric empirical band covered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch. On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-sided empirical index requires more calibration points than are often available. An infinite interval covers by construction. On rows where both are finite, the original wins on coverage *and* width. The change was reversed.

An overclaimed guarantee. The artifact was described as carrying a coverage guarantee without qualification. Checking the method against the conformal literature rather than against our own earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected in §8.

Provenance discipline. Two parser fixes were informed by inspecting rows that were already in the test set. Rather than silently retain them, we record which fixes were test-informed and exclude the affected family from the strict headline.

Mechanical verification. The figures this paper reports as results are macros generated from a claims registry that recomputes each value from source artifacts, and the accompanying code fails a test if an outward-facing document contains an untraced numeric literal. The exceptions are historical figures in the error list above (row counts and the -68pp delta), which are quoted from the provenance record rather than recomputed.

10 Conclusion

Per-model prediction of quantization accuracy damage does not work on published data, and the reason is largely measurement noise rather than model capacity. What does work is narrower: a per-scheme calibrated envelope, validated at 90.1% coverage on unseen checkpoints for 4 of the 6 schemes, and that says which of its own cells it cannot judge. Each cell is trusted; or insufficient

evidence, in which case the interval is still shown but flagged and the scheme demoted; or refused, in which case the interval is withheld. Today 9 of 17 cells are insufficient evidence and 0 are refused. The more useful contribution may be the limitation itself—the published record of quantized models is outcome-truncated, and we measured, on deliberately faulted recipes for models up to 1.5B, losses more than four times worse than anything it contains. A risk estimate built on that record is a floor on risk, not a ceiling, and should say so.

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